Algorithm 1 Region-Based Multicast Routing in the Local Tree

Require: Destination R_d, router region origin (X, Y), level $l \in \{1, 2, 3\}$, ingress d_{in}

Ensure: $\mathbf{D} = [D_P, D_{00}, D_{10}, D_{01}, D_{11}]$

$\mathbf{D} \leftarrow \mathbf{0}$; $s \leftarrow 2^{l-1}$

$R_v \leftarrow [X, X + 2s - 1] \times [Y, Y + 2s - 1]$

$R_{00} \leftarrow [X, X + s - 1] \times [Y, Y + s - 1]$

$R_{10} \leftarrow [X + s, X + 2s - 1] \times [Y, Y + s - 1]$

$R_{01} \leftarrow [X, X + s - 1] \times [Y + s, Y + 2s - 1]$

$R_{11} \leftarrow [X + s, X + 2s - 1] \times [Y + s, Y + 2s - 1]$

for $q \in \{00, 10, 01, 11\}$ **do**

  if $R_d \cap R_q \neq \emptyset \wedge (l = 1 \vee d_{\text{in}} \neq q)$ **then**

    $D_q \leftarrow 1$

  end if

end for

if $R_d \not\subseteq R_v \wedge d_{\text{in}} \neq P$ **then**

  $D_P \leftarrow 1$

end if

return D